

ICS 6B F26 Take Home Exam 1

Due: October 2, 2026 at 11:59PM

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UCI NetID :

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(alpha-numeric; NOT your student ID)

- Read the instructions offeach question careffily.
- Show your work. Where applicable, partial credit will be given on the work shown.
- Please keep in mind the academic honesty guidelines. This take-home exam is to be completed individually, with no outside help. You may use any resources ffrom our class (ZyBooks and resources ffrom Canvas), but you may not use any other online resources.
- You may choose to print the exam or use a digital editor ffor completing the exam. It is required that you use this PDF to complete your work. If you have no access to a printer or digital tools to ffulfill the exam, feel ffree to reach out to the staffs regarding your concerns.
- If you have any questions, please post a private Ed or attend available Off ce Hours. Note that we are not allowed to provide speciffc help to answering the exam questions.
- There are **7** problems on this test.
- There are a total **22** offpoints.
- Good luck!



Seat

Problem 1 (3 points)

Complete the truth table below for the following expressions:

- $(a \wedge b) \vee (\neg a \wedge \neg b)$
- $(a \leftrightarrow b) \wedge (c \vee \neg c)$

Row number	a	b	c	$\neg a \wedge \neg b$	$a \leftrightarrow b$	$(a \wedge b) \vee (\neg a \wedge \neg b)$	$(a \leftrightarrow b) \wedge (c \vee \neg c)$
1	T	T	T	F	T	T	T
2	T	T	F	F	T	T	T
3	T	F	T	F	F	F	F
4	T	F	F	F	F	F	F
5	F	T	T	F	F	F	F
6	F	T	F	F	F	F	F
7	F	F	T	T	T	T	T
8	F	F	F	T	T	T	T

If the above expressions are not equivalent then give the row number that proves that these expressions are not equivalent. If there is more than one correct answer then only give a single answer. If they are equivalent write 0 as your solution.

0

Problem 2 (2 points)

Consider the following two conditional statements. Convert them into the regular "If p , then q " form.

- Discrete structures exist unless Platonic forms are not real.

If Platonic forms are real

then, Discrete structures exist

- The sum of two odd numbers is odd only when Christmas is on Sunday.

If Christmas is on Sunday

then, The sum of two odd numbers is odd

Problem 3 (4 points)

Consider the following propositions.

- A: Octa likes hot dogs.
- B: Yousef practices his pitching.
- C: The ballpark is open.

Convert the following two sentences into logic according to the propositions above and prove that one sentence is the other sentence **negated**. Use DeMorgan's Law to convert the **negation** of the logical expression for sentence 1 into a logical expression for sentence 2 (use any additional laws as needed, and label all applications of laws).

- (Yousef does not practice his pitching if Octa likes hot dogs) and the ballpark is open.
- (Octa likes hot dogs and Yousef practices his pitching) or the ballpark is closed.

Logical expression for sentence 1:

$$(A \rightarrow \neg B) \wedge C$$

Logical expression for sentence 2:

$$(A \wedge B) \vee \neg C$$

Starting from negation of expr 1:

$$\neg((A \rightarrow \neg B) \wedge C)$$

$$\neg(A \rightarrow \neg B) \vee \neg C$$

$$(\neg(\neg A) \wedge \neg(\neg B)) \vee \neg C$$

$$(A \wedge B) \vee \neg C$$

Proof:

Thus proven.

Problem 4 (2 points)

Mark each of the following as Tautology, Contradiction or Neither. No explanation needed.

1. $(a \wedge \neg a) \rightarrow a$
☒ Tautology ☐ Contradiction ☐ Neither
2. $(a \vee b) \rightarrow (a \wedge b)$
☐ Tautology ☐ Contradiction ☒ Neither
3. $(b \rightarrow a) \rightarrow (a \rightarrow b)$
☐ Tautology ☐ Contradiction ☒ Neither
4. $(a \wedge b) \leftrightarrow (\neg a \wedge \neg b)$
☐ Tautology ☐ Contradiction ☒ Neither

Problem 5 (3 points)

Construct a logical expression which is only false when x, y and z are true but a, b and c are false. Prove using truth table or describe using a few words why your solution is true. Write the logical expression in the first box and justify your expression in the second box.

$$\neg(x \wedge y \wedge z) \vee (a \vee b \vee c)$$

Left side $\neg(x \wedge y \wedge z)$ is false if and only if x, y , and z are all true.

Right side $(a \vee b \vee c)$ is false if and only if a, b , and c are all false.

Therefore, the entire expression is false only in the single case where x, y, z are all true and a, b, c are all false.

For every other combination, at least one side is true.

Problem 6 (5 points)

Show that these statements are inconsistent, in other words, show the conjunction of these statements is a contradiction using laws of the logic. Use as many rows as you need (you could combine the steps if they are using the same law) but the expression must end with False (F).

1. If Billy does not practice the guitar, then he has time to play Counter Strike. ($\neg a \rightarrow b$)
2. If Billy has time to play Counter Strike, then he has free time. ($b \rightarrow c$)
3. If Billy likes to grind then he does not have free time. ($d \rightarrow \neg c$)
4. Billy does not practice the guitar and likes to grind. ($\neg a \wedge d$)

Expressions	Laws of Logic Used
$(\neg a \rightarrow b) \wedge (b \rightarrow c) \wedge (d \rightarrow \neg c) \wedge (\neg a \wedge d)$	Given
$(a' \rightarrow b) * (b \rightarrow c) * (d \rightarrow c') * (a' * d)$	Rewrite in easier form to type
$(a'' + b) * (b' + c) * (d' + c') * (a' * d)$	Material implication law
$(a + b) * (b' + c) * (d' + c') * (a' * d)$	Double Negation law
$a' * (a + b) * d * (d' + c') * (b' + c)$	Associative law
$((a' * a) + (a' * b)) * ((d * d') + (d * c')) * (b' + c)$	Distributive law
$(F + (a' * b)) * (F + (d * c')) * (b' + c)$	Complement law
$a' * b * d * c' * (b' + c)$	Identity law
$a' * d * ((b * c' * b') + (b * c' * c))$	Distributive law
$a' * d * ((F * c) + (F * b))$	Complement law
$a' * d * (F + F)$	Domination law
$a' * d * F$	Idempotent law
F	Domination law

Problem 7 (3 points)

Select all choices that are propositions.

- ☒ Max Planck invented quantum mechanics.
- ☐ This is not a proposition.
- ☒ If it is Monday, then I am the president of Tunisia.
- ☐ Tell me the day of the week it is today.
- ☒ The square of an odd number is odd.
- ☒ 15 is a prime number.
- ☐ Is 17 a prime number too?